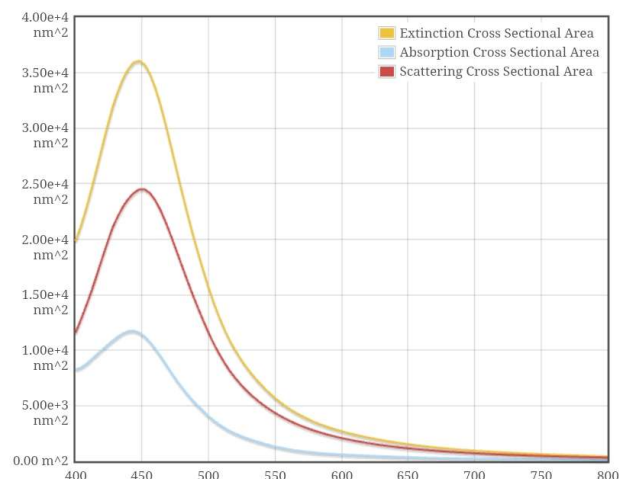


Note: Calculating optical response in variable media such as buffers, polymer matrices, biological media, coatings, or resins is useful for nanoparticle-enabled technologies across diagnostic, nanomedicine, and optical engineering applications.



Core/Shell Refractive Indices (RI) only need an input when Core/Shell Type is Other

Warning: your browser is set to not accept cookies. This site is optimized for better performance with cookies; if cookies are not enabled, you will not receive the optimal experience.

Core Type

Core RI

Core Diameter (nm)

Gold

20

Shell Type

Shell RI

Shell Thickness (nm)

Silver

25

Start λ (nm)

End λ (nm)

Medium RI

400

800

1.33

Plot!

Download Data

How to cite the nanoComposix Mie theory calculator

The Mie theory implementation used here was developed by Steve Oldenburg, Founder of nanoComposix, as part of his doctoral work at Rice University. More details on theory and code can be found in his dissertation, which can be cited as Oldenburg, S. J. *Light Scattering from Gold Nanoshells*. Diss. Rice University, 2000.

Materials commonly modeled with Mie theory

This calculator is designed for spherical single-component and core-shell particles. Use it to compare optical behavior across material classes, then explore which nanoComposix materials might be right for your application!

Gold Nanospheres

Compare size- and media-dependent extinction for gold spheres.

Silver Nanospheres

Model how silver sphere diameter influences peak position, broadening, absorption, and scattering.

Gold Nanoshells

Explore how silica core diameter and gold shell thickness tune absorption and scattering.

Silica-Shelled Nanospheres

Compare how a dielectric shell alters the calculated response of a plasmonic core.

Silica & Titania

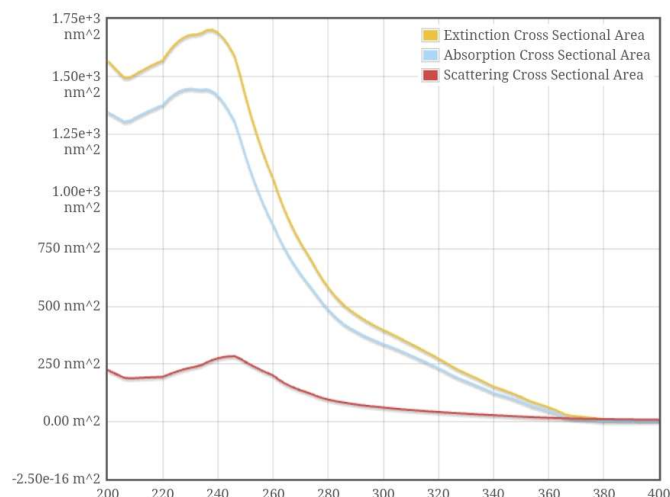
Evaluate size-dependent scattering and absorbance of dielectric particles.

Custom Nanoparticles

Tailor particle size and composition for your specialized optical requirements.

cross-section units in the downloaded data are in units of $\text{m}^2/\text{particle}$.

Note: Calculating optical response in variable media such as buffers, polymer matrices, biological media, coatings, or resins is useful for nanoparticle-enabled technologies across diagnostic, nanomedicine, and optical engineering applications.



Core/Shell Refractive Indices (RI) only need an input when Core/Shell Type is Other

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Core Type

TiO₂

Shell Type

None

Start λ (nm)

200

Core RI

Shell RI

End λ (nm)

400

Core Diameter (nm)

30

Shell Thickness (nm)

0

Medium RI

1.33

Plot!

Download Data

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